

10th grade

Learning evidence 2.1 Decision analysis supplementary solutions

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Problems

Problem 1: Problem description

A weekend coffee cart must choose a service plan for a street market. Plan A is a specialty menu and Plan B is a classic menu. Customer turnout can be high or low with probabilities 0,55 and 0,45. Profit outcomes are given directly: under high turnout Plan A earns 700 and Plan B earns 480, and under low turnout Plan A earns 250 and Plan B earns 200. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

C2

Decision Alternatives	Plan A (specialty menu), Plan B (classic menu)
States of Nature	Turnout high or low with probabilities 0,55 and 0,45
Events	Realized customer turnout in the market day
Consequences	Profit in dollars after revenue and costs

C3

Payoff is the profit in dollars given for each alternative and state.

State of nature	Probability	Plan A	Plan B
High turnout	0.55	700	480
Low turnout	0.45	250	200

C4

$$\text{máx}(\text{Plan A}) = \text{máx}\{700, 250\} = 700,$$

$$\text{máx}(\text{Plan B}) = \text{máx}\{480, 200\} = 480.$$

$$\text{máx}\{\text{máx}(\text{Plan A}), \text{máx}(\text{Plan B})\} = \text{máx}\{700, 480\} = 700.$$

The Maximax choice is Plan A with 700. The criterion is *risky* because it focuses only on the best possible payoff and ignores how likely it is.

C5

$$\min(\text{Plan A}) = \min\{700, 250\} = 250,$$

$$\min(\text{Plan B}) = \min\{480, 200\} = 200.$$

$$\max\{\min(\text{Plan A}), \min(\text{Plan B})\} = \max\{250, 200\} = 250.$$

The Maximin choice is Plan A with 250 because it has the least severe worst case. This is *safe* because it protects against the worst outcome.

C1

$$EV_A = 0,55(700) + 0,45(250) = 385 + 112,5 = 497,5,$$

$$EV_B = 0,55(480) + 0,45(200) = 264 + 90 = 354.$$

The expected value criterion chooses Plan A because 497,5 is the largest expected profit. This aligns with the Maximax and Maximin outcomes for Plan A, so it is the balanced decision using the probabilities.

Problem 2: Problem description

A campus bookstore must choose a stocking plan for a semester. Plan A is premium notebooks, Plan B is standard notebooks, and Plan C is discount notebooks. Demand for notebooks can be strong, moderate, or weak with probabilities 0,35, 0,45, and 0,20. Expected unit sales depend on demand and plan. Under strong demand Plan A sells 420 units, Plan B sells 520 units, and Plan C sells 600 units. Under moderate demand Plan A sells 300 units, Plan B sells 380 units, and Plan C sells 450 units. Under weak demand Plan A sells 180 units, Plan B sells 240 units, and Plan C sells 300 units. Unit prices are 22, 18, and 15 for Plans A, B, and C. Variable costs per unit are 10, 8, and 6. Use these parameters to calculate the profit payoff table. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

C2

Decision Alternatives	Plan A (premium), Plan B (standard), Plan C (discount)
States of Nature	Demand strong, moderate, weak with probabilities 0,35, 0,45, 0,20
Events	Realized demand level during the semester
Consequences	Profit in dollars after revenue and costs

C3

Payoff = revenue – cost, where revenue = (units × price) and cost = (units × variable cost).

<i>Units by plan and demand</i>			
Plan	Strong	Moderate	Weak
Plan A	420	300	180
Plan B	520	380	240
Plan C	600	450	300

Price parameters by plan

Plan	Price per unit
Plan A	22
Plan B	18
Plan C	15

Cost parameters by plan

Plan	Variable cost
Plan A	10
Plan B	8
Plan C	6

Revenue by demand

State of nature	Plan A revenue	Plan B revenue	Plan C revenue
Strong demand	$420 \cdot 22$ = 9240	$520 \cdot 18$ = 9360	$600 \cdot 15$ = 9000
Moderate demand	$300 \cdot 22$ = 6600	$380 \cdot 18$ = 6840	$450 \cdot 15$ = 6750
Weak demand	$180 \cdot 22$ = 3960	$240 \cdot 18$ = 4320	$300 \cdot 15$ = 4500

Cost by demand

State of nature	Plan A cost	Plan B cost	Plan C cost
Strong demand	$420 \cdot 10$ = 4200	$520 \cdot 8$ = 4160	$600 \cdot 6$ = 3600
Moderate demand	$300 \cdot 10$ = 3000	$380 \cdot 8$ = 3040	$450 \cdot 6$ = 2700
Weak demand	$180 \cdot 10$ = 1800	$240 \cdot 8$ = 1920	$300 \cdot 6$ = 1800

Profit is computed as Revenue minus Cost for each alternative and state.

State of nature	Probability	Plan A	Plan B	Plan C
Strong demand	0.35	$9240 - 4200$ = 5040	$9360 - 4160$ = 5200	$9000 - 3600$ = 5400
Moderate demand	0.45	$6600 - 3000$ = 3600	$6840 - 3040$ = 3800	$6750 - 2700$ = 4050
Weak demand	0.20	$3960 - 1800$ = 2160	$4320 - 1920$ = 2400	$4500 - 1800$ = 2700

C4

$$\text{máx(Plan A)} = \text{máx}\{5040, 3600, 2160\} = 5040,$$

$$\text{máx(Plan B)} = \text{máx}\{5200, 3800, 2400\} = 5200,$$

$$\text{máx(Plan C)} = \text{máx}\{5400, 4050, 2700\} = 5400.$$

$$\text{máx}\{\text{máx(Plan A)}, \text{máx(Plan B)}, \text{máx(Plan C)}\} = \text{máx}\{5040, 5200, 5400\} = 5400.$$

The Maximax choice is Plan C with 5400. The criterion is *risky* because it focuses only on the best possible payoff and ignores how likely it is.

C5

$$\text{mín(Plan A)} = \text{mín}\{5040, 3600, 2160\} = 2160,$$

$$\text{mín(Plan B)} = \text{mín}\{5200, 3800, 2400\} = 2400,$$

$$\text{mín(Plan C)} = \text{mín}\{5400, 4050, 2700\} = 2700.$$

$$\text{máx}\{\text{mín(Plan A)}, \text{mín(Plan B)}, \text{mín(Plan C)}\} = \text{máx}\{2160, 2400, 2700\} = 2700.$$

The Maximin choice is Plan C with 2700 because it gives the best worst case outcome. This is *safe* because it prioritizes the least harmful result.

C1

$$EV_A = 0,35(5040) + 0,45(3600) + 0,20(2160) = 1764 + 1620 + 432 = 3816,$$

$$EV_B = 0,35(5200) + 0,45(3800) + 0,20(2400) = 1820 + 1710 + 480 = 4010,$$

$$EV_C = 0,35(5400) + 0,45(4050) + 0,20(2700) = 1890 + 1822,5 + 540 = 4252,5.$$

The expected value criterion chooses Plan C because 4252,5 is the largest expected profit. This aligns with the Maximax and Maximin outcomes for Plan C, so it is the balanced decision using the probabilities.

Problem 3: Problem description

A rideshare cooperative must choose a service focus for the next season. Strategy A is an urban focus and Strategy B is a suburban focus. Demand can be strong or weak with probabilities 0,65 and 0,35. Energy costs can be low or high with probabilities 0,55 and 0,45, and these costs are independent of demand. Revenue in thousands of dollars depends on demand and strategy. Under strong demand revenue is 520 for Strategy A and 560 for Strategy B. Under weak demand revenue is 260 for Strategy A and 240 for Strategy B. Operating costs in thousands of dollars depend on energy costs and strategy. Under low costs operating costs are 200 for Strategy A and 230 for Strategy B. Under high costs operating costs are 260 for Strategy A and 300 for Strategy B. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

C2

Decision Alternatives	Strategy A (urban), Strategy B (suburban)
States of Nature	Demand strong or weak with energy costs low or high
Events	Realized demand level paired with energy cost conditions
Consequences	Profit in thousands of dollars from revenue minus operating costs

C3

Payoff = revenue – cost, where revenue is determined by demand and cost is determined by energy prices.

Revenue parameters by strategy

Strategy	Strong	Weak
Probabilities	0.65	0.35
Strategy A	520	260
Strategy B	560	240

Cost parameters by strategy

Strategy	Low cost	High cost
Probabilities	0.55	0.45
Strategy A	200	260
Strategy B	230	300

Profit is computed as Revenue minus Cost for each alternative and state.

Profit parameters by strategy

State of nature	Probability	Strategy A	Strategy B
Strong demand, low cost	$0,65 \cdot 0,55$ $= 0,3575$	$520 - 200$ $= 320$	$560 - 230$ $= 330$
Strong demand, high cost	$0,65 \cdot 0,45$ $= 0,2925$	$520 - 260$ $= 260$	$560 - 300$ $= 260$
Weak demand, low cost	$0,35 \cdot 0,55$ $= 0,1925$	$260 - 200$ $= 60$	$240 - 230$ $= 10$
Weak demand, high cost	$0,35 \cdot 0,45$ $= 0,1575$	$260 - 260$ $= 0$	$240 - 300$ $= -60$

Payoff table

State of nature	Probability	Strategy A	Strategy B
Strong demand, low cost	0.3575	320	330
Strong demand, high cost	0.2925	260	260
Weak demand, low cost	0.1925	60	10
Weak demand, high cost	0.1575	0	-60

C4

$$\text{máx}(\text{Strategy A}) = \text{máx}\{320, 260, 60, 0\} = 320,$$

$$\text{máx}(\text{Strategy B}) = \text{máx}\{330, 260, 10, -60\} = 330.$$

$$\text{máx}\{\text{máx}(\text{Strategy A}), \text{máx}(\text{Strategy B})\} = \text{máx}\{320, 330\} = 330.$$

The Maximax choice is Strategy B with 330. The criterion is *risky* because it focuses only on the best possible payoff and ignores how likely it is.

C5

$$\text{mín}(\text{Strategy A}) = \text{mín}\{320, 260, 60, 0\} = 0,$$

$$\text{mín}(\text{Strategy B}) = \text{mín}\{330, 260, 10, -60\} = -60.$$

$$\text{máx}\{\text{mín}(\text{Strategy A}), \text{mín}(\text{Strategy B})\} = \text{máx}\{0, -60\} = 0.$$

The Maximin choice is Strategy A with 0 because it has the least severe worst case. This is *safe* because it protects against the worst outcome.

C1

Expected values use the combined state probabilities.

$$\begin{aligned}EV_A &= 0,3575(320) + 0,2925(260) + 0,1925(60) + 0,1575(0) \\&= 114,4 + 76,05 + 11,55 + 0 = 202,0, \\EV_B &= 0,3575(330) + 0,2925(260) + 0,1925(10) + 0,1575(-60) \\&= 117,975 + 76,05 + 1,925 - 9,45 = 186,5.\end{aligned}$$

The expected value criterion chooses Strategy A because 202,0 is the largest expected profit. This differs from the Maximax choice and matches the Maximin choice, so the expected value emphasizes weighted outcomes.

Problem 4: Problem description

A regional utility must choose a generation mix for the next year. Strategy A is solar heavy, Strategy B is wind heavy, and Strategy C is balanced. Demand can be high, medium, or low with probabilities 0,25, 0,50, and 0,25. Operating cost conditions can be low or high with probabilities 0,60 and 0,40, and these conditions are independent of demand. Revenue in thousands of dollars depends on demand and strategy. Under high demand revenue is 680 for Strategy A, 620 for Strategy B, and 650 for Strategy C. Under medium demand revenue is 530 for Strategy A, 500 for Strategy B, and 515 for Strategy C. Under low demand revenue is 360 for Strategy A, 380 for Strategy B, and 370 for Strategy C. Operating costs in thousands of dollars depend on cost conditions and strategy. Under low costs operating costs are 300 for Strategy A, 280 for Strategy B, and 290 for Strategy C. Under high costs operating costs are 380 for Strategy A, 360 for Strategy B, and 370 for Strategy C. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

C2

Decision Alternatives	Strategy A (solar), Strategy B (wind), Strategy C (balanced)
States of Nature	Demand high, medium, low with cost conditions low or high
Events	Realized demand level paired with operating cost conditions
Consequences	Profit in thousands of dollars from revenue minus operating costs

C3

Payoff = revenue – cost, where revenue is determined by demand and cost is determined by operating conditions.

Revenue parameters by strategy

Strategy	High	Medium	Low
Probabilities	0.25	0.50	0.25
Strategy A	680	530	360
Strategy B	620	500	380
Strategy C	650	515	370

Cost parameters by strategy

Strategy	Low cost	High cost
Probabilities	0.60	0.40
Strategy A	300	380
Strategy B	280	360
Strategy C	290	370

Profit is computed as Revenue minus Cost for each alternative and state.

<i>Profit parameters by strategy</i>				
State of nature	Probability	Strategy A	Strategy B	Strategy C
High demand, low cost	$0,25 \cdot 0,60$	$680 - 300$	$620 - 280$	$650 - 290$
	$= 0,15$	$= 380$	$= 340$	$= 360$
High demand, high cost	$0,25 \cdot 0,40$	$680 - 380$	$620 - 360$	$650 - 370$
	$= 0,10$	$= 300$	$= 260$	$= 280$
Medium demand, low cost	$0,50 \cdot 0,60$	$530 - 300$	$500 - 280$	$515 - 290$
	$= 0,30$	$= 230$	$= 220$	$= 225$
Medium demand, high cost	$0,50 \cdot 0,40$	$530 - 380$	$500 - 360$	$515 - 370$
	$= 0,20$	$= 150$	$= 140$	$= 145$
Low demand, low cost	$0,25 \cdot 0,60$	$360 - 300$	$380 - 280$	$370 - 290$
	$= 0,15$	$= 60$	$= 100$	$= 80$
Low demand, high cost	$0,25 \cdot 0,40$	$360 - 380$	$380 - 360$	$370 - 370$
	$= 0,10$	$= -20$	$= 20$	$= 0$

<i>Payoff table</i>				
State of nature	Probability	Strategy A	Strategy B	Strategy C
High demand, low cost	0.15	380	340	360
High demand, high cost	0.10	300	260	280
Medium demand, low cost	0.30	230	220	225
Medium demand, high cost	0.20	150	140	145
Low demand, low cost	0.15	60	100	80
Low demand, high cost	0.10	-20	20	0

C4

$$\text{máx}(\text{Strategy A}) = \text{máx}\{380, 300, 230, 150, 60, -20\} = 380,$$

$$\text{máx}(\text{Strategy B}) = \text{máx}\{340, 260, 220, 140, 100, 20\} = 340,$$

$$\text{máx}(\text{Strategy C}) = \text{máx}\{360, 280, 225, 145, 80, 0\} = 360.$$

$$\text{máx}\{\text{máx}(\text{Strategy A}), \text{máx}(\text{Strategy B}), \text{máx}(\text{Strategy C})\} = \text{máx}\{380, 340, 360\} = 380.$$

The Maximax choice is Strategy A with 380. The criterion is *risky* because it focuses only on the best possible payoff and ignores how likely it is.

C5

$$\text{mín}(\text{Strategy A}) = \text{mín}\{380, 300, 230, 150, 60, -20\} = -20,$$

$$\text{mín}(\text{Strategy B}) = \text{mín}\{340, 260, 220, 140, 100, 20\} = 20,$$

$$\text{mín}(\text{Strategy C}) = \text{mín}\{360, 280, 225, 145, 80, 0\} = 0.$$

$$\text{máx}\{\text{mín}(\text{Strategy A}), \text{mín}(\text{Strategy B}), \text{mín}(\text{Strategy C})\} = \text{máx}\{-20, 20, 0\} = 20.$$

The Maximin choice is Strategy B with 20 because it has the least severe worst case. This is *safe* because it protects against the worst outcome.

C1

Expected values use the combined state probabilities.

$$\begin{aligned}
 EV_A &= 0,15(380) + 0,10(300) + 0,30(230) + 0,20(150) + 0,15(60) + 0,10(-20) \\
 &= 57 + 30 + 69 + 30 + 9 - 2 = 193, \\
 EV_B &= 0,15(340) + 0,10(260) + 0,30(220) + 0,20(140) + 0,15(100) + 0,10(20) \\
 &= 51 + 26 + 66 + 28 + 15 + 2 = 188, \\
 EV_C &= 0,15(360) + 0,10(280) + 0,30(225) + 0,20(145) + 0,15(80) + 0,10(0) \\
 &= 54 + 28 + 67,5 + 29 + 12 + 0 = 190,5.
 \end{aligned}$$

The expected value criterion chooses Strategy A because 193 is the largest expected profit. This differs from the Maximin choice and matches the Maximax choice, so the expected value gives a balanced decision using the probabilities.

Problem 5: Problem description

A community fitness hub must choose a membership model for the next year. Model A is a weekday focus, Model B is a family pass, and Model C is an evening pass. Demand can be high, medium, or low with probabilities 0,40, 0,35, and 0,25. Staffing costs can be favorable or unfavorable with probabilities 0,55 and 0,45, and these costs are independent of demand. Expected membership counts depend on demand and model. Under high demand Model A has 700 members, Model B has 650 members, and Model C has 760 members. Under medium demand Model A has 520 members, Model B has 480 members, and Model C has 600 members. Under low demand Model A has 360 members, Model B has 320 members, and Model C has 420 members. Membership fees are 38 for Model A, 42 for Model B, and 34 for Model C per member per year. Variable staffing cost per member is 20 when costs are favorable and 28 when costs are unfavorable. Fixed annual costs are 8200 for Model A, 9600 for Model B, and 7000 for Model C. Which decision is the risky choice under Maximax, the safe choice under Maximin, and the balanced choice under Expected Value?

C2

Decision Alternatives	Model A (weekday), Model B (family), Model C (evening)
States of Nature	Demand high, medium, low with staffing costs favorable or unfavorable
Events	Realized demand level paired with staffing cost conditions
Consequences	Annual profit in dollars from membership fees minus staffing and fixed costs

C3

Payoff = revenue – cost, where revenue = (members × fee) and cost = (members × variable cost) + fixed cost.

<i>Members per model by demand</i>			
Model	High	Medium	Low
Model A	700	520	360
Model B	650	480	320
Model C	760	600	420

Cost parameters by model

Model	Favorable	Unfavorable	Fixed cost
Model A	20	28	8200
Model B	20	28	9600
Model C	20	28	7000

Revenue parameters by member

Model	Fee per member
Model A	38
Model B	42
Model C	34

Revenue by demand

State of nature	Model A revenue	Model B revenue	Model C revenue
High demand	$700 \cdot 38$	$650 \cdot 42$	$760 \cdot 34$
	= 26600	= 27300	= 25840
Medium demand	$520 \cdot 38$	$480 \cdot 42$	$600 \cdot 34$
	= 19760	= 20160	= 20400
Low demand	$360 \cdot 38$	$320 \cdot 42$	$420 \cdot 34$
	= 13680	= 13440	= 14280

Cost by demand and staffing condition

State of nature	Model A cost	Model B cost	Model C cost
High demand, favorable cost	$700 \cdot 20 + 8200$	$650 \cdot 20 + 9600$	$760 \cdot 20 + 7000$
	= 22200	= 22600	= 22200
High demand, unfavorable cost	$700 \cdot 28 + 8200$	$650 \cdot 28 + 9600$	$760 \cdot 28 + 7000$
	= 27800	= 27800	= 28280
Medium demand, favorable cost	$520 \cdot 20 + 8200$	$480 \cdot 20 + 9600$	$600 \cdot 20 + 7000$
	= 18600	= 19200	= 19000
Medium demand, unfavorable cost	$520 \cdot 28 + 8200$	$480 \cdot 28 + 9600$	$600 \cdot 28 + 7000$
	= 22760	= 23040	= 23800
Low demand, favorable cost	$360 \cdot 20 + 8200$	$320 \cdot 20 + 9600$	$420 \cdot 20 + 7000$
	= 15400	= 16000	= 15400
Low demand, unfavorable cost	$360 \cdot 28 + 8200$	$320 \cdot 28 + 9600$	$420 \cdot 28 + 7000$
	= 18280	= 18560	= 18760

Profit is computed as Revenue minus Cost for each alternative and state.

Profit parameters by model

State of nature	Model A	Model B	Model C	Probabilities
High demand, favorable cost	26600 – 22200 = 4400	27300 – 22600 = 4700	25840 – 22200 = 3640	$0,40 \cdot 0,55$ = 0,22
High demand, unfavorable cost	26600 – 27800 = –1200	27300 – 27800 = –500	25840 – 28280 = –2440	$0,40 \cdot 0,45$ = 0,18
Medium demand, favorable cost	19760 – 18600 = 1160	20160 – 19200 = 960	20400 – 19000 = 1400	$0,35 \cdot 0,55$ = 0,1925
Medium demand, unfavorable cost	19760 – 22760 = –3000	20160 – 23040 = –2880	20400 – 23800 = –3400	$0,35 \cdot 0,45$ = 0,1575
Low demand, favorable cost	13680 – 15400 = –1720	13440 – 16000 = –2560	14280 – 15400 = –1120	$0,25 \cdot 0,55$ = 0,1375
Low demand, unfavorable cost	13680 – 18280 = –4600	13440 – 18560 = –5120	14280 – 18760 = –4480	$0,25 \cdot 0,45$ = 0,1125

Combined probabilities are $0,40 \cdot 0,55 = 0,22$, $0,40 \cdot 0,45 = 0,18$, $0,35 \cdot 0,55 = 0,1925$, $0,35 \cdot 0,45 = 0,1575$, $0,25 \cdot 0,55 = 0,1375$, $0,25 \cdot 0,45 = 0,1125$.

State of nature	Probability	Model A	Model B	Model C
High demand, favorable cost	0.22	4400	4700	3640
High demand, unfavorable cost	0.18	-1200	-500	-2440
Medium demand, favorable cost	0.1925	1160	960	1400
Medium demand, unfavorable cost	0.1575	-3000	-2880	-3400
Low demand, favorable cost	0.1375	-1720	-2560	-1120
Low demand, unfavorable cost	0.1125	-4600	-5120	-4480

C4

$$\max(\text{Model A}) = \max\{4400, -1200, 1160, -3000, -1720, -4600\} = 4400,$$

$$\max(\text{Model B}) = \max\{4700, -500, 960, -2880, -2560, -5120\} = 4700,$$

$$\max(\text{Model C}) = \max\{3640, -2440, 1400, -3400, -1120, -4480\} = 3640.$$

$$\max\{\max(\text{Model A}), \max(\text{Model B}), \max(\text{Model C})\} = \max\{4400, 4700, 3640\} = 4700.$$

The Maximax choice is Model B with 4700. The criterion is *risky* because it focuses only on the best possible payoff and ignores how likely it is.

C5

$$\min(\text{Model A}) = \min\{4400, -1200, 1160, -3000, -1720, -4600\} = -4600,$$

$$\min(\text{Model B}) = \min\{4700, -500, 960, -2880, -2560, -5120\} = -5120,$$

$$\min(\text{Model C}) = \min\{3640, -2440, 1400, -3400, -1120, -4480\} = -4480.$$

$$\max\{\min(\text{Model A}), \min(\text{Model B}), \min(\text{Model C})\} = \max\{-4600, -5120, -4480\} = -4480.$$

The Maximin choice is Model C with -4480 because it has the least severe worst case loss. This is *safe* because it protects against the worst outcome.

C1

Expected values use the combined state probabilities.

$$\begin{aligned}EV_A &= 0,22(4400) + 0,18(-1200) + 0,1925(1160) + 0,1575(-3000) + 0,1375(-1720) + 0,1125(-4600) \\ &= 968 - 216 + 223,3 - 472,5 - 236,5 - 517,5 = -251,2,\end{aligned}$$

$$\begin{aligned}EV_B &= 0,22(4700) + 0,18(-500) + 0,1925(960) + 0,1575(-2880) + 0,1375(-2560) + 0,1125(-5120) \\ &= 1034 - 90 + 184,8 - 453,6 - 352 - 576 = -252,8,\end{aligned}$$

$$\begin{aligned}EV_C &= 0,22(3640) + 0,18(-2440) + 0,1925(1400) + 0,1575(-3400) + 0,1375(-1120) + 0,1125(-4480) \\ &= 800,8 - 439,2 + 269,5 - 535,5 - 154 - 504 = -562,4.\end{aligned}$$

The expected value criterion chooses Model A because $-251,2$ is the largest expected profit. This differs from the Maximin and Maximax choices, so the expected value gives a balanced decision using the probabilities.